

PROBLEM SPACE

AI TRIAGE SUPPORT SYSTEM CO-DESIGN CANVAS

Setting B — HRI: Mercer Observation Unit Robotic Kiosk

USER

Group(s)	Characteristics	Needs
- Observation Unit nurses responsible for reviewing robot-assisted vital sign assessments and responding to alerts. - Healthcare assistants supporting routine observations and patient guidance. primary users	- Clinically trained <i>What are the users like?</i> - Responsible for several patients at once - Need fast, reliable information - May not be beside the kiosk during every assessment - Work in a busy, noisy environment - Must interpret robot-generated readings alongside clinical judgement.	- Reliable collection of vital signs <i>What are needed?</i> - Immediate notification of abnormal or missing readings - Clear indication when a sensor fails - Manual fallback when the kiosk cannot complete an assessment - Easy access to the patient's results & clear indication when nurse review is required before any clinical action.
secondary users - Patients, Porters, Observation Unit physicians - Biomedical engineering staff responsible for maintaining sensors and hardware. - Clinical IT staff - Hospital governance personnel	Patients may vary in age, mobility, language and health literacy. Some may be anxious, confused or unfamiliar with technology. The technical staff needs clear fault and maintenance information and the governance staff need auditability and evidence of safe operation. Some patients may require assistance using the kiosk because of mobility, vision or cognitive limitations.	- Simple instructions - Clear indication of what the robot is doing - Ability to request human help - Privacy during assessment - Reassurance that the robot supports staff rather than replacing them - Accessible interaction for patients with different physical and cognitive abilities.
- Collect routine observations safely - Identify abnormal readings quickly - Reduce repetitive monitoring workload - Receive alerts when an assessment is incomplete. - Improve the efficiency of routine observation routine reducing patient safety. primary users	Goal(s) - Improve consistency of Observation Unit monitoring - Support timely clinical escalation - Maintain nurse oversight of all robot-assisted assessments	
secondary users - Complete a guided vital-sign check safely - Understand each step of the interaction - Obtain help easily when needed SHORT-TERM	- Improve access to timely monitoring - Maintain patient autonomy, privacy and confidence - Reduce delays without reducing human care - Increase patient confidence in robot-assisted observations through clear communication. LONG-TERM	

ROBOT

Task(s)			
• Guide the patient through the assessment • Collect vital signs from connected sensors • Check whether readings are complete and valid • Display simple instructions • Transmit validated observations to the nurse dashboard • Alert staff if readings are abnormal or incomplete • Request manual assistance when a sensor fails		• Support repeated observation checks • Integrate with the hospital Electronic Health Record and observation systems. • Maintain logs of readings, faults and staff responses • Adapt instructions for accessibility and patient needs • Support safe escalation without making final clinical decisions.	
SHORT-TERM		LONG-TERM	
Advantages What tasks does the AI system perform?			
• Uses calm, simple prompts • Reassures patients during the assessment • Directs patients to request help when needed • Uses clear visual prompts alongside spoken or written instructions.	• Reduces uncertainty by explaining each step • Uses non-threatening language and movement • Avoids presenting the robot as a replacement for staff & maintains a calm, predictable interaction throughout the assessment.	• Adjustable language and text size • Can repeat instructions • Can adapt pace for elderly or anxious patients	• Repeats measurements consistently • Checks whether readings are complete • Records time and device status accurately • Requests repeat measurements automatically when readings are incomplete.
Social skills	User's emotional response	Personalization	Precise tasks
• Collects blood pressure, pulse, oxygen saturation and temperature • Flags missing or unreliable readings • Transfers data to staff for review • Automatically timestamps every observation.	• Ideally remains stationary beside the vitals station • Does not move close to patients unless necessary • Uses a stable base to reduce collision risk	• Very limited or none • Does not physically move patients or equipment • May adjust screen angle or sensor position only • Never performs physical clinical interventions.	• Sends validated observations directly to the nurse dashboard. • Stores fault and interaction logs • Supports future EHR integration • Provides manual entry fallback when connectivity fails
Data collection with sensors	Mobility	Environment manipulation	Connection to systems